#6 Aplique el método de Gauss - Jordan para determinar el conjunto solución del sistema:

(4 Pts)

$$\begin{cases} x+y+z+w=5 \\ 2x+3y+2z+2w=13 \\ 3x+4y+3z+3w=18 \end{cases}$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 5 \\ 2 & 3 & 2 & 2 & 13 \\ 3 & 4 & 3 & 3 & 18 \end{array}\right)$$

$$\begin{array}{r} \\ -2\cdot F1+\widetilde{F2} \\ -3\cdot F1+\widetilde{F3} \end{array} \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 5 \\ 0 & 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & 3 \end{array}\right)$$

$$\begin{array}{rc} & \begin{array}{ccccc} x & y & z & w & S \end{array} \\ \begin{array}{r} -F2+\widetilde{F1} \\ \\ -F2+\widetilde{F3} \end{array} & \left(\begin{array}{cccc|c} 1 & 0 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right) \end{array}$$

$$0 \quad 0 \quad 0 \quad 0 \mid 0$$

$$x+z+w=2 \rightarrow x=2-z-w$$

$$y=3 \qquad\qquad y=3$$

$$\boxed{R/\ S=\{(2-z-w,\ 3,\ z,\ w)\}}$$

#7 Considere matrices A, B y X cuadradas de orden 2×2. Se sabe que $(A+B)^T$ es invertible y que satisfacen la siguiente igualdad:

$$XA^T = 2I - (BX^T)^T.$$

$(A+B)^T \cdot X \to$

a) Utilice propiedades matriciales para verificar que $X = 2\left(A^T + B^T\right)^{-1}$.

$A \cdot B = C$

$B = C \cdot A^{-1}$; A es invertible

$A \cdot B \neq B \cdot A$

$X \cdot A^T = 2I - (BX^T)^T$

$X A^T = 2I - (X^T)^T \cdot B^T$ $\qquad (A \cdot B)^T = B^T \cdot A^T$

$X \cdot A^T = 2I - X \cdot B^T$ $\qquad (X^T)^T = X$

$X A^T + X \cdot B^T = 2I$ $\qquad XA + XB = X(A+B)$

$AX + BX = (A+B)X$

$X(A^T + B^T) = 2I$

$X(A+B)^T = 2I$ $\qquad (A+B)^T = A^T + B^T$

$X = 2I[(A+B)^T]^{-1}$ $\qquad AI = A$

$$\boxed{X = 2(A^T + B^T)^{-1}}$$

$2X + AX$

$(2I + A)X$

$\hookrightarrow$ ojo!!!

$$X = 2\left(\underbrace{A^T + B^T}\right)^{-1}$$

b) Usando el resultado de la parte *a*) y si se cumple que

$$A + B = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix},$$

calcule explícitamente la matriz X.

$$A^T + B^T = (A+B)^T$$

$$(A+B)^T = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}$$

$$\left[(A+B)^T\right]^{-1} \qquad \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 5 & 0 & 1 \end{array}\right)$$

$$-3 \cdot F1 + F2 \quad \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -1 & -3 & 1 \end{array}\right)$$

$$-1 \cdot F2 \quad \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & 3 & -1 \end{array}\right)$$

$$-2 \cdot F2 + F1 \quad \left(\begin{array}{cc|cc} 1 & 0 & -5 & 2 \\ 0 & 1 & 3 & -1 \end{array}\right)$$

$$\left[(A+B)^T\right]^{-1} = \begin{pmatrix} 5 & 2 \\ 3 & -1 \end{pmatrix}$$

$$2\left(A^T + B^T\right)^{-1} = 2 \cdot \begin{pmatrix} 5 & 2 \\ 3 & -1 \end{pmatrix}$$

$$\boxed{\begin{pmatrix} 10 & 4 \\ 6 & -2 \end{pmatrix}}$$

$$A = \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix}, \quad I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$A \cdot I = \begin{pmatrix} 2+3\cdot 0 & 3 \\ 4 & 5 \end{pmatrix}$$

$$A = \begin{pmatrix} 2 & 2 & 1 \\ 0 & 4 & 5 \\ 3 & 1 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 4 & 5 & 6 \\ 3 & 2 & 1 \\ 4 & 0 & 1 \end{pmatrix}$$

$$A \cdot B = 2\cdot 4 + 2\cdot 3 + 1\cdot 4$$

NO es $\pm\frac{\pi}{2}$ v $\pm\pi$

Arg $\to$ $\frac{b}{a} = \tan(\theta)$

$-\frac{\pi}{2}$	$\frac{\pi}{2}$	$-\pi$	π
$a=0$	$a=0$	$b=0$	$b=0$
$b<0$	$b>0$	$a>0$	$a<0$

#4 Sea $z = a + bi$ un número complejo que cumple las siguientes condiciones:

$$\begin{cases} |z+3-3i| = \sqrt{74} \\ \operatorname{Arg}(z+4i) = \pi. \end{cases} \to b=0$$

$z = a + bi$

Determine todos los números z que cumplan, de manera simultánea, ambas condiciones.

$a + bi + 4i$ $\qquad b = 0$

$(a) + (b+4)i$ $\qquad a < 0$

$b + 4 = 0$ $\qquad a < 0$

$b = -4$

$|z+3-3i| = \sqrt{74}$

$|a+bi+3-3i| = \sqrt{74}$

$|(a+3)+(b-3)i| = \sqrt{74}$ $\qquad b=-4$

$(a+3)^2 + (b-3)^2 = (\sqrt{74})^2$

$a^2 + 6a + 3^2 + (-4-3)^2 = 74$

$a^2+6a+3^2+(-4-3)^2=74$

$a^2+6a+9+49-74=0$

$a^2+6a-16=0$

$$\begin{array}{lll} a & 8 & =8a \\ a & -2 & =-2a \\ \hline & & 6a \end{array}$$

$(a+8)(a-2)=0$

$a=-8 \quad \cancel{a=2} \qquad a<0$

$a=-8 \quad b=-4$

$\boxed{-8-4i}$

$\ln(e)^x = x$

$\ln(a \cdot b) = \ln(a) + \ln(b)$

$\ln(x)^n = n\ln(x)$

$r = \sqrt{a^2+b^2}$

$\theta = \arctan\left(\frac{b}{a}\right)$

$a + bi \to$ Rectangular

$r\,\text{cis}(\theta) \to$ Polar

$r \cdot e^{\theta i} \to$ Exponencial

$r\,\text{cis}(\theta) \to r e^{\theta i}$

$a < 0 \quad b > 0 \to +\pi$

$a < 0 \quad b < 0 \to -\pi$

$a = 0 \quad b > 0 \to \frac{\pi}{2}$

$a = 0 \quad b < 0 \to -\frac{\pi}{2}$

$i,\ i+1,\ i-1$

Rectangular

8. Exprese $(-1-i)^i$ en la forma $a+bi$, con a y b números reales.

$z = (-1-i)^i$

A exponencial

$\ln(z) = \ln(-1-i)^i$

$r = \sqrt{(-1)^2+(-1)^2} = \sqrt{2}$

$= i\ln(-1-i)$

$\theta = \tan^{-1}\left(\frac{-1}{-1}\right) - \pi$

$= \frac{-3\pi}{4}$

$\sqrt{2} \cdot e^{\frac{-3\pi}{4} \cdot i}$ $\quad r \cdot e^{\theta i}$

$i\ln\left(\sqrt{2} \cdot e^{\frac{-3\pi}{4} \cdot i}\right)$

$\ln(a \cdot b) = \ln(a) + \ln(b)$

$i\ln(\sqrt{2}) + i\cancel{\ln}\left(\cancel{e}\right)^{\frac{-3\pi}{4} i}$

$\ln(\sqrt{2})i + \frac{-3\pi}{4} i^2$

$$\ln(\sqrt{2})i + -\frac{3\pi}{4} i^2$$

$$e^{\ln(\sqrt{2})i + \frac{3\pi}{4}}$$

$$e^{\ln(\sqrt{2})i} \cdot e^{\frac{3\pi}{4}} \qquad e^{\theta i} = \operatorname{cis}(\theta)$$

$$\operatorname{cis}(\ln(\sqrt{2}))\, e^{\frac{3\pi}{4}}$$

$$[\cos(\ln(\sqrt{2})) + \operatorname{sen}(\ln(\sqrt{2}))i] \cdot e^{\frac{3\pi}{4}}$$

$$\cos(\ln(\sqrt{2})) \cdot e^{\frac{3\pi}{4}} + \operatorname{sen}(\ln(\sqrt{4})) \cdot e^{\frac{3\pi}{4}} i$$

$$\boxed{9{,}92 + 3{,}58 i}$$

5. Considere las matrices A y X de dimensiones 3×3 y la matriz B de dimensión 2×3, para las cuales está definida la igualdad

$$XA + B^tB = 2X.$$

$2I - A$ es invertible

a) Utilice álgebra matricial para despejar X de la ecuación anterior. (3 puntos)

$$XA + B^T \cdot B = 2X$$

$$B^T \cdot B = 2X - XA \qquad K \cdot A = A \cdot K$$

$$B^T \cdot B = X2 - XA$$

$$B^T \cdot B = X(2I - A)$$

$$\boxed{X = B^T \cdot B\,(2I - A)^{-1}}$$

6. Determine el número complejo z que satisface simultáneamente las condiciones siguientes: (4 puntos)

$z = a + bi$

$$\begin{cases} |z-1| = 2 \\ \operatorname{Arg}(z-2-i) = -\frac{\pi}{2}. \end{cases} \longrightarrow \pm\frac{\pi}{2}$$

$a + bi - 2 - i \qquad a = 0 \quad b < 0$

$(a-2)(b-1)i$

$a - 2 = 0 \qquad b - 1 < 0$

$a = 2 \qquad b < 1$

$|z-1| = 2$

$|a + bi - 1| = 2$

$|(a-1) + (b)i| = 2$

$(a-1)^2 + b^2 = 2^2$

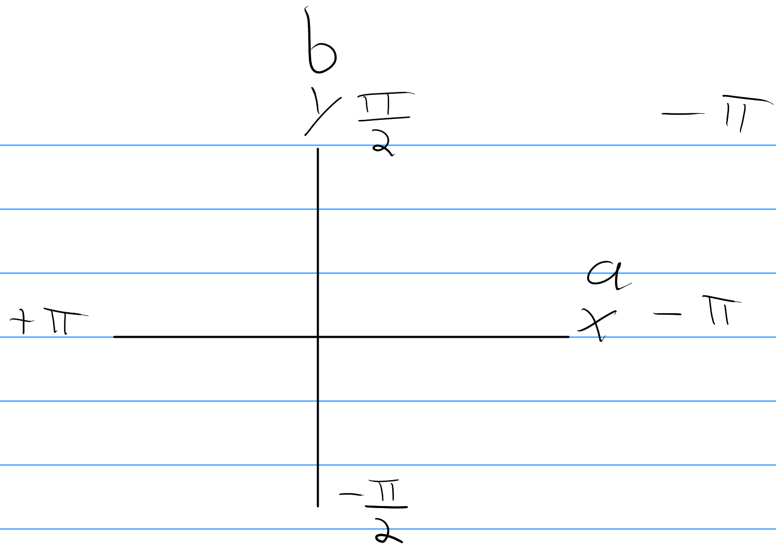
$(2-1)^2 + b^2 = 4$

$b^2 = 3$

$b = \pm\sqrt{3} \qquad b < 0$

$b = -\sqrt{3}$

$\boxed{\text{R/}\ \ 2 - \sqrt{3}\,i}$



$b=0 \quad a>0$

$b=0 \quad a<0$

14. Encuentre el o los números $z \in \mathbb{C}$ que satisfacen simultáneamente las siguientes condiciones: R/ $z = -3\sqrt{2}i$

$\bar{z} = a - bi$

$$\begin{cases} |z - \bar{z}| = 6\sqrt{2} \\ \arg(\bar{z} - 3\sqrt{2}) = \dfrac{3\pi}{4} \end{cases}$$

$$a - bi - 3\sqrt{2} = \frac{3\pi}{4}$$

$$(a - 3\sqrt{2}) + (-b)i = \frac{3\pi}{4} \qquad \frac{3\pi}{4} \cdot \frac{180}{\pi}$$

135°

$a - 3\sqrt{2} < 0 \qquad -b > 0$

$a < 3\sqrt{2} \qquad b < 0 \qquad 90 < 135 < 180$

$$(a - 3\sqrt{2}) + (-b)i = \frac{3\pi}{4} \qquad a < 0 \quad b > 0$$

$$\frac{-b}{a - 3\sqrt{2}} = \tan\left(\frac{3\pi}{4}\right)$$

$$\frac{\cancel{-}b}{a - 3\sqrt{2}} = \cancel{-}1$$

$$b = a - 3\sqrt{2}$$

$$a = b + 3\sqrt{2}$$

$$|z - \bar{z}| = 6\sqrt{2}$$

$$|a + bi - (a - bi)| = 6\sqrt{2}$$

$$|\cancel{a} + bi \cancel{- a} + bi| = 6\sqrt{2}$$

$$(2b)i = 6\sqrt{2}$$

$$4b^2 = (6\sqrt{2})^2$$

$$4b^2 = 72$$

$$b^2 = 18$$

$$b = \pm 3\sqrt{2} \qquad b < 0 \qquad a < 0$$

$$b = 3\sqrt{2}$$

$$\boxed{RL - 3\sqrt{2}\, i}$$